UNIVERSITY SCHOLARSHIP ALLOCATION

Python Activity-Based Learning | Project Documentation

Project overview
A beginner-friendly Python program that collects a student's marks, attendance, and family income and prints whether the student meets the scholarship rules.

1. Problem Statement

Design a simple system to evaluate a student's scholarship eligibility from five subject marks, attendance percentage, and family income. The program calculates the aggregate percentage and grants eligibility only when all three rules are satisfied.

2. Objectives

- Ask for the student's name and five subject marks.
- Calculate the total marks and aggregate percentage.
- Check the aggregate, attendance, and family-income requirements together.
- Display a clear eligible or not-eligible result using the student's name.

Eligibility Rules

Aggregate percentage	At least 60% out of 500 total marks
Attendance	At least 75%
Family income	Less than 100,000

Scope clarification: Subject marks are added to calculate the aggregate. This version does not apply a separate minimum score to each subject; eligibility follows the aggregate, attendance, and income rules listed above.

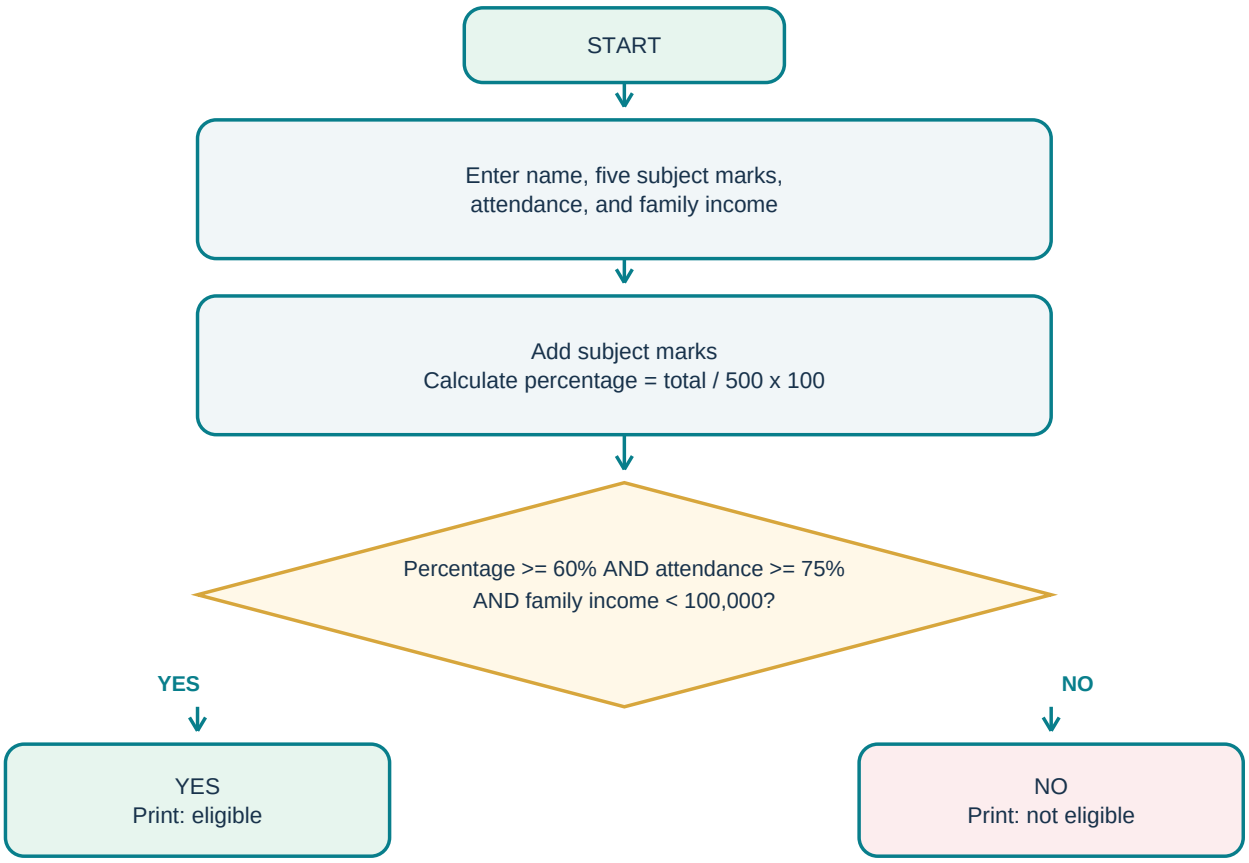
3. Assumptions

- There are five subjects: Python, Maths, Physics, English, and C Programming.
- Each subject is marked out of 100, so the maximum aggregate is 500.
- The expected marks and attendance are entered as numbers from 0 to 100.
- Family income is entered as a number in the same units as the 100,000 limit.
- Input is assumed to be valid numeric data; the program does not validate ranges or non-numeric entries.
- A student must satisfy all three rules. Income equal to 100,000 does not qualify.

4. Algorithm

1. Read the student's name.
2. Read marks for Python, Maths, Physics, English, and C Programming.
3. Read attendance percentage and family income.
4. Add all five subject marks to get total marks.
5. Calculate aggregate percentage as (total marks / 500) x 100.
6. If aggregate percentage is at least 60, attendance is at least 75, and family income is below 100,000, print that the student is eligible.
7. Otherwise, print that the student is not eligible.

5. Flowchart



6. Pseudocode

```
BEGIN
  INPUT student_name
  INPUT python_marks, maths_marks, physics_marks
  INPUT english_marks, c_programming_marks
  INPUT attendance, family_income
  total_marks = sum of the five subject marks
  marks_percentage = total_marks / 500 * 100
  IF marks_percentage >= 60 AND attendance >= 75
    AND family_income < 100000 THEN
    DISPLAY student_name + ' is eligible for scholarship'
  ELSE
    DISPLAY student_name + ' is not eligible for the scholarship'
  END IF
END
```

7. Source Code

```
student_name = input("Enter student name: ")

python_marks = float(input("Enter Python marks: "))
maths_marks = float(input("Enter Maths marks: "))
physics_marks = float(input("Enter Physics marks: "))
english_marks = float(input("Enter English marks: "))
c_programming_marks = float(input("Enter C Programming marks: "))

attendance = float(input("Enter attendance %: "))
family_income = float(input("Enter family income: "))

total_marks = (
    python_marks
    + maths_marks
    + physics_marks
    + english_marks
    + c_programming_marks
)
marks_percentage = total_marks / 500 * 100

if marks_percentage >= 60 and attendance >= 75 and family_income < 100000:
    print(student_name, "is eligible for scholarship")
else:
    print(student_name, "is not eligible for the scholarship")
```

The listing above is taken from the project source file **University_scholarship_allocation.py**.

8. Test Cases

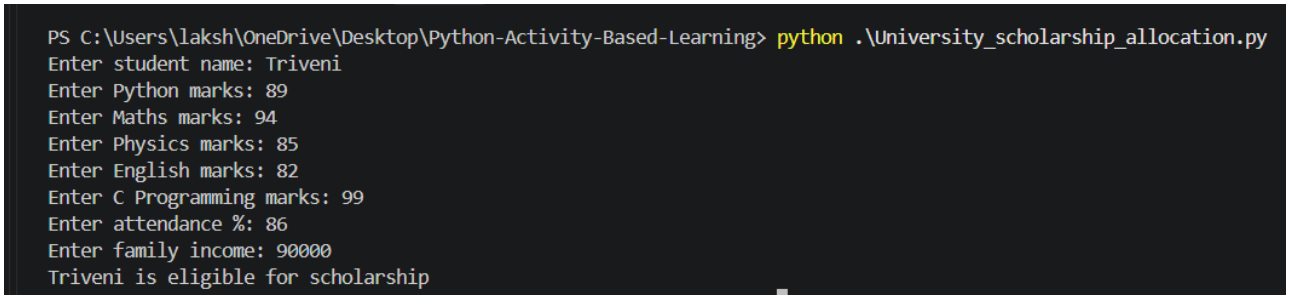
The first three cases are the supplied terminal screenshots. Additional cases check the exact thresholds and the strict income limit.

Case	Marks (total / %)	Attendance Income	Expected result / reason
1	449 / 500 89.8%	86% 90,000	Eligible (Triveni); all rules pass.
2	460 / 500 92%	83% 150,000	Not eligible (Ramya); income is over the limit.
3	465 / 500 93%	74% 87,000	Not eligible (Haritha); attendance is below 75%.
4	474 / 500 94.8%	98% 100,000	Not eligible; income must be strictly below 100,000.
5	300 / 500 60%	75% 99,999	Eligible; aggregate and attendance meet their boundaries.
6	299 / 500 59.8%	80% 90,000	Not eligible; aggregate is below 60%.

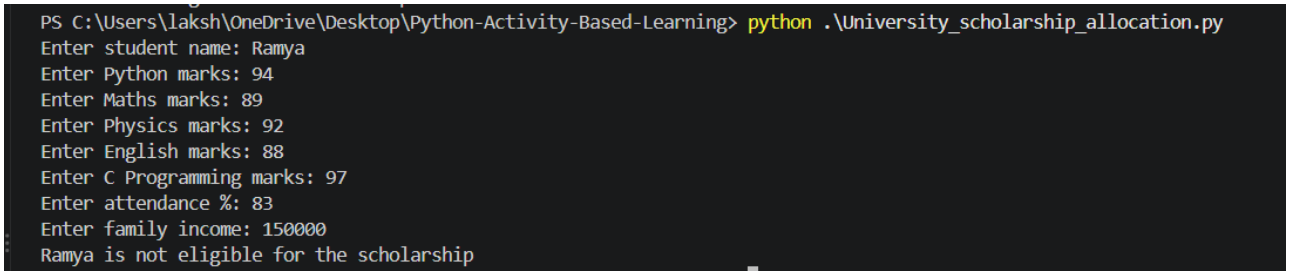
Verification: All six listed cases were run against the Python program. Their observed outputs matched the expected eligibility result.

9. Output Screenshots

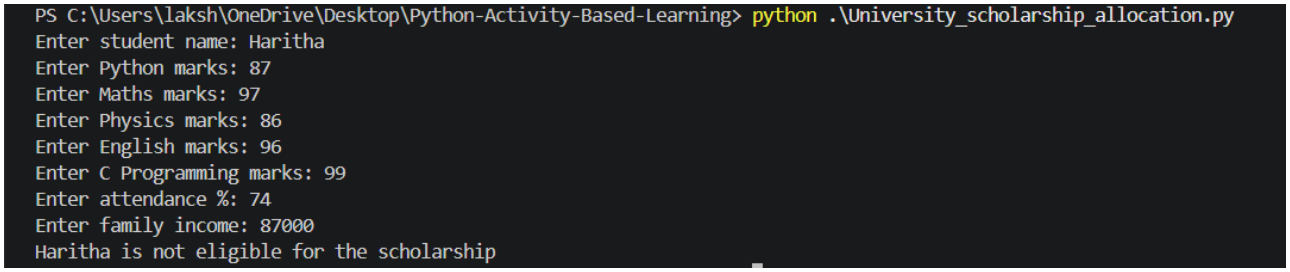
These screenshots show the program's actual console output for an eligible student and for two different reasons for ineligibility.



Test Case 1 - Triveni meets the aggregate, attendance, and income requirements.



Test Case 2 - Ramya does not qualify because family income is above the limit.



Test Case 3 - Haritha does not qualify because attendance is below 75%.

10. Conclusion

The program demonstrates a basic decision-making task in Python using input, arithmetic, comparison operators, and an if-else statement. It reports eligibility only when the aggregate percentage is at least 60%, attendance is at least 75%, and family income is less than 100,000. Testing covered the provided examples and the important threshold boundaries.